Lecture 4

Irreducibility, averaging, and Schur's lemma

Lecture 3 ended with a representation that split into two smaller ones. This lecture asks when that happens and what the pieces are. A representation is irreducible when it has no smaller representation inside it. Averaging an inner product over the group makes every representation of a finite group unitary, and from unitarity follows Maschke's theorem: every representation is a direct sum of irreducible ones. Schur's lemma then says how rigid the irreducible pieces are. Only multiples of the identity commute with an irreducible representation, and that is the reason symmetric Hamiltonians have degenerate levels.

By the end of this lecture you will be able to:

- Find invariant subspaces and test irreducibility.
- Make a representation unitary by averaging, and split it by Maschke's theorem.
- Apply Schur's lemma.

4.1 Invariant subspaces and irreducibility

The last example of Lecture 3 displayed a representation containing smaller ones: the rotations of C_3 preserve the lines $\mathbb{C}v_+$ and $\mathbb{C}v_-$. The general notions:

Definition 4.1 (Invariant subspace; irreducibility)**.** Let (ρ, V) be a representation. A subspace $W \subseteq V$ is *invariant* if

$$\rho(g)W \subseteq W \quad \text{for all } g \in G.$$

Equivalently, $\rho(g)W = W$ (it suffices to act on the condition above with $\rho(g^{-1})$). An invariant subspace carries the *subrepresentation*

$$\rho|_W : g \mapsto \rho(g)|_W.$$

The representation is *irreducible* if $\dim V \geq 1$ and its only invariant subspaces are $\{0\}$ and V; otherwise it is *reducible*. Every one-dimensional representation is irreducible.

A line $\mathbb{C}v$ is invariant exactly when v is an eigenvector of every $\rho(g)$ at once. So a two-dimensional representation is reducible exactly when its operators have a common eigenvector.

Example 4.2 (Inside the permutation representation)**.** In the permutation representation of S_n on $\mathbb{C}^n$ (Lecture 3), the line

$$L = \mathbb{C}\,u_0, \qquad u_0 = e_1 + e_2 + \cdots + e_n,$$

is invariant: permuting the summands leaves u_0 fixed, so L carries a copy of the trivial representation. A second invariant subspace is

$$V_0 = \left\{ v \in \mathbb{C}^n : \textstyle\sum_i v_i = 0 \right\},$$

of dimension $n - 1$: permuting the components of v does not change their sum. So the permutation representation is reducible for every $n \geq 2$, and $\mathbb{C}^n = L \oplus V_0$ as vector spaces, both summands invariant. The subrepresentation on V_0 is called the *standard representation* of S_n; for $n = 3$ the tutorial proves it irreducible. ◁

In the example, the invariant subspace came with an invariant complement. That this always happens for finite groups is Maschke's theorem, and it is a genuine theorem: for the infinite group $(\mathbb{Z}, +)$ it fails, as the self-study example at the end of Section 4.3 shows. The proof goes through unitarity.

4.2 Unitarity: the averaging trick

Equip V with a Hermitian inner product $\langle \cdot, \cdot \rangle$, conjugate-linear in the first (left) argument (the physics convention). An operator U is *unitary* if $\langle Uu, Uv \rangle = \langle u, v \rangle$ for all $u, v \in V$, and a representation is *unitary* (with respect to the given product) if every $\rho(g)$ is. In quantum mechanics this is the natural demand: a symmetry must not change transition probabilities.

Theorem 4.3 (Unitarisability)**.** Let (ρ, V) be a representation of a *finite* group G, and let $\langle \cdot, \cdot \rangle_0$ be any inner product on V. Then

$$\langle u, v \rangle \;:=\; \frac{1}{|G|} \sum_{h \in G} \langle \rho(h)u,\ \rho(h)v \rangle_0$$

is an inner product on V with respect to which ρ is unitary.

Proof. Sesquilinearity (conjugate-linearity in the first argument and linearity in the second) is inherited term by term, and positivity holds because a sum of nonnegative terms vanishes only if each term does — in particular the $h = e$ term $\langle u, u \rangle_0$, so $\langle u, u \rangle = 0$ forces $u = 0$.

For unitarity, fix g and compute

$$\langle \rho(g)u, \rho(g)v \rangle = \frac{1}{|G|} \sum_{h \in G} \langle \rho(hg)u, \rho(hg)v \rangle_0 = \frac{1}{|G|} \sum_{h' \in G} \langle \rho(h')u, \rho(h')v \rangle_0 = \langle u, v \rangle\,,$$

because as h runs once over G, so does $h' = hg$: the map $h \mapsto hg$ is a bijection of G (cancellation — the Latin-square property of the Cayley table, Lecture 1). ∎

Remark 4.4 (The averaging trick)**.** *Average over the group, and the average is invariant, because translating the summation variable only reshuffles the sum.* This is the most reused idea of the subject: it gives the orthogonality of characters in Lecture 5, and in Part II, with the sum replaced by an integral, it carries the whole theory.

The theorem changes the inner product and leaves the operators alone. In a basis orthonormal for the new product, the matrices $D(g)$ are unitary matrices in the ordinary sense. So *every matrix representation of a finite group is equivalent to one by unitary matrices*, and from here on we may and do assume the matrices unitary whenever that helps. The change of basis can be found explicitly, as follows.

Unitarising a matrix representation

Fix a basis and let M be the Gram matrix of the invariant product, $M_{ij} = \langle e_i, e_j \rangle$, so that

$$\langle u, v \rangle = \sum_{i,j=1}^{n} \bar{u}_i \, v_j \, \langle e_i, e_j \rangle = \sum_{i,j=1}^{n} \bar{u}_i \, M_{ij} \, v_j = u^{\dagger} M v.$$

Then M is Hermitian and positive definite. Since $\langle \rho(g)u, \rho(g)v \rangle = u^{\dagger} D(g)^{\dagger} M D(g)\, v$, invariance of the product, $\langle \rho(g)u, \rho(g)v \rangle = \langle u, v \rangle$ for all $u, v \in V$, says exactly

$$D(g)^{\dagger} M D(g) = M \qquad \text{for all } g \in G.$$

Corollary 4.5 (Unitarising a matrix representation)**.** Let D be a matrix representation of a finite group G and let M be as above. Let S be any invertible matrix with

$$S^{\dagger} M S = \mathbb{1},$$

that is, any matrix whose columns are a basis orthonormal for the invariant product. Then

$$U(g) := S^{-1} D(g)\, S$$

is unitary for every $g \in G$, and U is a matrix representation equivalent to D.

Proof. Rearranging $S^{\dagger} M S = \mathbb{1}$ gives $M = (S^{-1})^{\dagger} S^{-1}$, so

$$U(g)^{\dagger} U(g) = S^{\dagger} D(g)^{\dagger} (S^{-1})^{\dagger} S^{-1} D(g)\, S = S^{\dagger} \big(D(g)^{\dagger} M D(g) \big) S = S^{\dagger} M S = \mathbb{1}.$$

That U is a representation, and equivalent to D, is the definition of equivalence in Lecture 3, with intertwiner S^{-1}. ∎

How to find S. Apply Gram–Schmidt to the standard basis $(e_1, \dots, e_n)$ in the invariant product and take the results as the columns of S. Because Gram–Schmidt builds the j-th vector out of the first j, this S is upper triangular, and $M = (S^{-1})^{\dagger} S^{-1}$ is the Cholesky factorisation of M. It is convenient to compute with a computer algebra system.

[tutorial] **Example 4.6** (Averaging, concretely)**.** Let $C_2 = \{e, a\}$ act on $\mathbb{C}^2$ by

$$D(e) = \mathbb{1}, \qquad D(a) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad D(a)^2 = \mathbb{1} \ \checkmark,$$

a representation that is *not* unitary for the standard product: $D(a)^{\dagger} D(a) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \neq \mathbb{1}$. Average the standard product $\langle u, v \rangle_0 = u^{\dagger} v$ over both group elements:

$$\langle u, v \rangle = \frac{1}{2} \Big(\langle D(e)u, D(e)v \rangle_0 + \langle D(a)u, D(a)v \rangle_0 \Big) = \frac{1}{2} \Big(u^{\dagger} v + u^{\dagger} D(a)^{\dagger} D(a)\, v \Big) = u^{\dagger} M v,$$

where

$$M = \frac{1}{2}\Big(\mathbb{1} + D(a)^{\dagger} D(a)\Big) = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}.$$

One verifies $D(a)^{\dagger} M D(a) = M$: the representation is unitary for the new (invariant) product.

The same thing is visible on passing to the eigenbasis of $D(a)$. The matrix $D(a)$ has two eigenvalues, $+1$ and -1, with eigenvectors $v_+ = \binom{1}{0}$ and $v_- = \binom{1}{-2}$. The vectors $v_\pm$ are orthogonal in the *invariant* product though not in the standard one:

$$v_+^{\dagger} M v_- = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 3/2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -5/2 \end{pmatrix} = 0.$$

In the basis (v_+, v_-) the matrix $D(a)$ becomes $\operatorname{diag}(1, -1)$ — manifestly unitary, and incidentally displaying the decomposition of this representation into the two one-dimensional representations of C_2.

That route used eigenvectors, and it worked only because this representation happens to split into one-dimensional pieces. The recipe above needs none: Gram–Schmidt (e_1, e_2) in the invariant product and its norm $\|v\| = \sqrt{\langle v, v\rangle}$:

$$f_1 = \frac{e_1}{\|e_1\|} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad f_2 = \frac{e_2 - \langle f_1, e_2\rangle\, f_1}{\|e_2 - \langle f_1, e_2\rangle\, f_1\|} = \frac{2}{\sqrt{5}} \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}.$$

Here we used

$$\langle f_1, e_2\rangle = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 3/2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{2} \quad \text{and} \quad e_2 - \langle f_1, e_2\rangle\, f_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix},$$

and the square of that vector's norm is

$$\begin{pmatrix} -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 1 & 1/2 \\ 1/2 & 3/2 \end{pmatrix} \begin{pmatrix} -1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 5/4 \end{pmatrix} = \frac{5}{4}.$$

Taking f_1 and f_2 as the columns, we obtain

$$S = \begin{pmatrix} | & | \\ f_1 & f_2 \\ | & | \end{pmatrix} = \begin{pmatrix} 1 & -1/\sqrt{5} \\ 0 & 2/\sqrt{5} \end{pmatrix} \quad \text{and} \quad S^{-1} D(a)\, S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the same unitary matrix as before: f_2 is v_- normalised in the invariant product, up to sign. The inverse of S

$$S^{-1} = \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{5}/2 \end{pmatrix}$$

is upper triangular and satisfies $M = (S^{-1})^{\dagger} S^{-1}$ — the Cholesky factor, as it must be. ◁

The structural consequence of unitarity is this:

> **Corollary 4.7.** In a unitary representation, the orthogonal complement of an invariant subspace is invariant.

Proof. Let W be invariant and $w' \in W^{\perp}$. For any $g \in G$ and $w \in W$,

$$\langle \rho(g) w',\, w\rangle = \big\langle w',\, \rho(g)^{-1} w\big\rangle = \big\langle w',\, \rho(g^{-1}) w\big\rangle = 0,$$

the first step by unitarity and the last because $\rho(g^{-1}) w \in W$. Hence $\rho(g) w' \in W^{\perp}$. ∎

4.3 Maschke's theorem: complete reducibility

Theorem 4.8 (Maschke)**.** Every finite-dimensional complex representation of a finite group is a direct sum of irreducible subrepresentations: there are invariant subspaces $W_1, \dots, W_k$, each irreducible, with

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_k.$$

Proof. Induct on $\dim V$. If $\dim V = 0$ the statement is empty (the direct sum of no representations), so assume $\dim V \geq 1$. If V is irreducible (in particular if $\dim V = 1$), there is nothing to prove.

Otherwise, equip V with an invariant inner product (Theorem 4.3) and let W be an invariant subspace with $0 \neq W \neq V$. By Corollary 4.7, $V = W \oplus W^{\perp}$ with both summands invariant and of strictly smaller dimension; apply the induction hypothesis to each. ∎

In matrix language: *there is a single invertible S such that*

$$S\,D(g)\,S^{-1} = \begin{pmatrix} D^{(i_1)}(g) & & \\ & D^{(i_2)}(g) & \\ & & \ddots \end{pmatrix} \qquad \text{for every } g \in G$$

simultaneously, with each diagonal block an irreducible matrix representation (Figure 4.1). The force of the statement lies in the word *simultaneously*: there is one basis in which every group element takes the same block form at the same time. Finding that basis for representations of physical interest is, in a practical sense, what Lectures 5 to 7 are about.

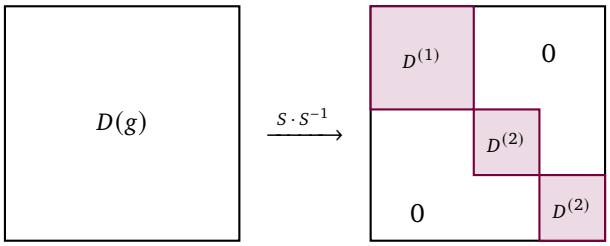


Figure 4.1: Maschke's theorem in matrix form: a single change of basis S brings every $D(g)$, for all $g \in G$ at once, to the same block-diagonal form with irreducible blocks. Blocks may repeat: here the irreducible representation $D^{(2)}$ occurs with multiplicity two.

Notation. Collecting equivalent blocks, we write

$$V \;\cong\; \bigoplus_i m_i\, V^{(i)},$$

where $V^{(i)}$ runs over the inequivalent irreducible representations of G, with $n_i = \dim V^{(i)}$, and the integer $m_i \geq 0$ is the *multiplicity* of $V^{(i)}$ in V. So $\dim V = \sum_i m_i n_i$. The subspaces W_j themselves are not unique (two copies of the trivial representation can be resliced into different pairs of invariant lines). That the multiplicities are unique is proved at the end of this lecture; Lecture 5 computes them without hunting for invariant subspaces.

What each hypothesis is for *[self-study]*

Both hypotheses of Maschke's theorem are necessary. First, finiteness of the group.

Example 4.9 (Reducible but indecomposable)**.** Let the *infinite* group $(\mathbb{Z},+)$ be represented on $\mathbb{C}^2$ by the operators whose matrices in the basis (e_1,e_2) are

$$D(n)=\begin{pmatrix}1 & n\\ 0 & 1\end{pmatrix}, \qquad D(n)D(m)=\begin{pmatrix}1 & n+m\\ 0 & 1\end{pmatrix}=D(n+m).$$

The line $\mathbb{C}e_1$ is invariant, so the representation is reducible. But no *complementary* invariant line exists: any line other than $\mathbb{C}e_1$ is spanned by some $\binom{a}{1}$, and

$$D(1)\begin{pmatrix}a\\1\end{pmatrix}=\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}\begin{pmatrix}a\\1\end{pmatrix}=\begin{pmatrix}a+1\\1\end{pmatrix}$$

is never proportional to $\binom{a}{1}$. The representation cannot be written as a direct sum of smaller ones — the matrices can be made simultaneously triangular but never simultaneously block-diagonal. There is no finite sum to average over, and no invariant inner product exists. ◁

Second, the field. Over fields of prime characteristic p dividing $|G|$, the averaging factor $1/|G|$ does not exist and complete reducibility fails; what survives in that case is the subject called modular representation theory. Over $\mathbb{R}$, by contrast, the proof goes through verbatim with real inner products — though, as the rotations of C_3 showed in Lecture 3, the *irreducible pieces* over $\mathbb{R}$ and over $\mathbb{C}$ can differ. In Part II, compactness of the Lie group will play the role finiteness plays here.

4.4 Worked example: the permutation representation of S_3 *[tutorial]*

We now execute the whole programme by hand on the three-dimensional permutation representation of S_3, with its invariant subspaces $L=\mathbb{C}u_0$ and $V_0=\{\sum_i v_i=0\}$ from Example 4.2. Since $(1\,2)$ and $(1\,2\,3)$ generate S_3, every identity we need can be checked on these two elements alone.

Step 1: adapt the basis. Choose

$$u_0=e_1+e_2+e_3, \qquad f_1=e_1-e_2, \qquad f_2=e_2-e_3,$$

so that (u_0,f_1,f_2) is a basis of $\mathbb{C}^3$ with u_0 spanning L and (f_1,f_2) spanning V_0.

Step 2: compute the blocks. The transposition $(1\,2)$ swaps e_1,e_2 and fixes e_3, so

$$\rho\big((1\,2)\big)f_1=e_2-e_1=-f_1, \qquad \rho\big((1\,2)\big)f_2=e_1-e_3=f_1+f_2.$$

The cycle $(1\,2\,3)$ sends $e_1\mapsto e_2\mapsto e_3\mapsto e_1$, so

$$\rho\big((1\,2\,3)\big)f_1=e_2-e_3=f_2, \qquad \rho\big((1\,2\,3)\big)f_2=e_3-e_1=-f_1-f_2.$$

Reading off columns, the matrices of the subrepresentation on V_0 in the basis (f_1,f_2) are

$$B\big((1\,2)\big)=\begin{pmatrix}-1 & 1\\ 0 & 1\end{pmatrix}, \qquad B\big((1\,2\,3)\big)=\begin{pmatrix}0 & -1\\ 1 & -1\end{pmatrix},$$

and indeed $B((1\,2))^2 = \mathbb{1}$, $B((1\,2\,3))^3 = \mathbb{1}$. In the full basis (u_0, f_1, f_2), with

$$S = \begin{pmatrix} | & | & | \\ u_0 & f_1 & f_2 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 1 & 0 & -1 \end{pmatrix}$$

the change-of-basis matrix, we obtain in block form

$$S^{-1} D(g)\, S = \begin{pmatrix} 1 & 0 \\ 0 & B(g) \end{pmatrix} \quad \text{for all } g \in S_3.$$

This is block-diagonal form, with blocks of sizes 1 and 2, as direct multiplication confirms.

Step 3: the 2×2 block is irreducible. A proper nonzero invariant subspace of V_0 would be a line, spanned by a common eigenvector of all the $B(g)$ — in particular of both generators. The eigenvectors of $B((1\,2))$ (eigenvalues -1 and $+1$) are the multiples of

$$w_- = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad w_+ = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

and neither line survives the cycle:

$$B\big((1\,2\,3)\big)w_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \nparallel w_-, \qquad B\big((1\,2\,3)\big)w_+ = \begin{pmatrix} -2 \\ -1 \end{pmatrix} \nparallel w_+.$$

No common eigenvector, no invariant line: the standard representation of S_3 is irreducible. Altogether

$$\mathbb{C}^3 \;\cong\; V^{\text{triv}} \oplus V^{\text{std}},$$

the trivial representation plus a two-dimensional irreducible one, each with multiplicity one — and the sign representation, a third irreducible, does not appear at all. That these three exhaust the irreducibles of S_3 is proved in Lecture 5, which also reads the multiplicities $1, 1, 0$ off in one line, with no hunting for invariant subspaces at all.

Remark 4.10 (The geometry of the standard representation)**.** The decomposition has a picture (Figure 4.2). S_3 permutes the coordinate axes of $\mathbb{R}^3$; the diagonal direction u_0 is fixed, and the action turns the perpendicular plane V_0 into itself. The orthogonal projections of e_1, e_2, e_3 onto that plane form an equilateral triangle, which the group permutes — so the standard representation is nothing but the defining representation of $D_3 \cong S_3$ (Lecture 1) in disguise: the abstract block B and the geometric matrices of Lecture 3 are equivalent. One group, the same irreducible two-dimensional representation, two coordinate descriptions.

4.5 Schur's lemma

Maschke's theorem breaks every representation into irreducible pieces. How precise is this decomposition? The answer comes from the maps between them. Recall from Lecture 3 that an *intertwiner* between representations (ρ_V, V) and (ρ_W, W) of G is a linear map $T\colon V \to W$ with

$$T\,\rho_V(g) = \rho_W(g)\,T \quad \text{for all } g \in G.$$

There we required T invertible and called the representations equivalent. Now we drop invertibility and study the vector space of *all* intertwiners, denoted $\mathrm{Hom}_G(V, W)$. The most important example in physics has $V = W$: an operator that commutes with every symmetry operation, such as a symmetric Hamiltonian. The decisive observation is one line long:

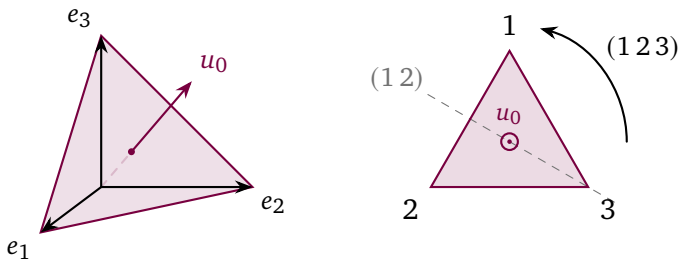


Figure 4.2: Left: the tips of e_1, e_2, e_3 span an equilateral triangle perpendicular to u_0. Right: its projection onto V_0, seen down u_0. The cycle $(1\,2\,3)$ rotates it through $2\pi/3$. The transposition $(1\,2)$ reflects it in the dashed axis.

Lemma 4.11. If $T \in \mathrm{Hom}_G(V, W)$, then $\ker T$ is an invariant subspace of V and $\mathrm{im}\, T$ is an invariant subspace of W.

Proof. If $Tv = 0$ then $T\rho_V(g)v = \rho_W(g)Tv = 0$, so $\rho_V(g)v \in \ker T$. And $\rho_W(g)\,Tv = T\rho_V(g)v \in \mathrm{im}\, T$. ∎

An *irreducible* representation has no invariant subspaces other than the zero subspace and the whole space. For irreducible representations the lemma therefore leaves only two possibilities.

Theorem 4.12 (Schur's lemma, first form)**.** Let V and W be irreducible representations of G and $T \in \mathrm{Hom}_G(V, W)$. Then either $T = 0$ or T is an isomorphism (so $V \cong W$). In particular, between *inequivalent* irreducible representations the only intertwiner is zero.

Proof. Suppose $T \neq 0$. By Lemma 4.11, $\ker T$ is an invariant subspace of the irreducible V, and $\ker T \neq V$ since $T \neq 0$; hence $\ker T = \{0\}$ and T is injective.

Likewise $\mathrm{im}\, T$ is a nonzero invariant subspace of the irreducible W, hence $\mathrm{im}\, T = W$. So T is surjective. ∎

Theorem 4.13 (Schur's lemma, second form)**.** Let (ρ, V) be a *finite-dimensional complex* irreducible representation, and let $T \in \mathrm{Hom}_G(V, V)$, i.e. $T\rho(g) = \rho(g)T$ for all $g \in G$. Then $T = \lambda\, \mathrm{id}_V$ for some $\lambda \in \mathbb{C}$:

$$\mathrm{Hom}_G(V, V) = \mathbb{C}\ \mathrm{id}_V\,.$$

Proof. Because $\mathbb{C}$ is algebraically closed, the characteristic polynomial of T has a root. So T has an eigenvalue $\lambda \in \mathbb{C}$.

Then $T - \lambda\, \mathrm{id}_V$ is again an intertwiner, since sums and scalar multiples of intertwiners intertwine, and id_V commutes with every $\rho(g)$. The kernel of $T - \lambda\, \mathrm{id}_V$ is the eigenspace of λ, so this kernel is nonzero.

By Theorem 4.12 a non-invertible intertwiner from V to V must vanish. Applied to $T - \lambda\, \mathrm{id}_V$, this gives $T = \lambda\, \mathrm{id}_V$. ∎

In words: the eigenspace of T is invariant and not $\{0\}$, and an irreducible representation has no invariant subspaces but $\{0\}$ and the whole space. So the eigenspace is everything.

Corollary 4.14. For complex irreducible representations V, W of a group G,

$$\dim \mathrm{Hom}_G(V,W) = \begin{cases} 1, & V \cong W, \\ 0, & V \not\cong W. \end{cases}$$

Proof. If $V \not\cong W$ this is the first form. If $V \cong W$, fix an isomorphism $T_0 \in \mathrm{Hom}_G(V,W)$; for any $T \in \mathrm{Hom}_G(V,W)$ the composite $T_0^{-1}T \in \mathrm{Hom}_G(V,V)$ is a scalar λ id by the second form, so $T = \lambda T_0$. ∎

Corollary 4.15 (Abelian groups)**.** Every complex irreducible representation of an abelian group is one-dimensional. In particular the n representations $\chi_k(r^m) = e^{2\pi \mathrm{i} km/n}$ of Lecture 3 are all the irreducible representations of C_n.

Proof. Let (ρ, V) be irreducible and G abelian. For fixed $h \in G$ the operator $\rho(h)$ commutes with every $\rho(g)$ — that is, $\rho(h) \in \mathrm{Hom}_G(V,V)$ — so $\rho(h) = \lambda_h \,\mathrm{id}_V$ by the second form. So every group element acts as a scalar. Then *every* subspace of V is invariant, and irreducibility forces $\dim V = 1$.

For $G = C_n$, a one-dimensional representation is fixed by the number $\rho(r)$, and $\rho(r)^n = \rho(r^n) = 1$ makes it an n-th root of unity, $\rho(r) = e^{2\pi \mathrm{i} k/n}$ for some k. That is χ_k. ∎

So the rotation representation of C_3 *had* to split over $\mathbb{C}$, as Lecture 3 found by hand. Over $\mathbb{R}$ the proof fails, for want of an eigenvalue; "irreducible" in this course means irreducible over $\mathbb{C}$.

Remark 4.16 (Schur's lemma over the real numbers)**.** The second form fails over $\mathbb{R}$, where eigenvalues need not exist. Let $C_4 = \langle r \rangle$ act irreducibly on $\mathbb{R}^2$ by the quarter-turn $J = R(\pi/2) = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The action commutes with every matrix

$$a\mathbb{1} + bJ = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \qquad a, b \in \mathbb{R}.$$

So the *commutant*, the space of all matrices commuting with the action, is two-dimensional. The commutant is a copy of the field $\mathbb{C}$ among the real 2×2 matrices, with J playing the role of i. Over $\mathbb{C}$ the representation splits into $\chi_1 \oplus \chi_{-1}$, and Schur's lemma holds again.

Read physically, the second form says that *an operator commuting with an irreducible symmetry action is a multiple of the identity*. A Hamiltonian commuting with the symmetry is therefore a single number on an irreducible representation that occurs once: one energy for the whole representation, so the level's degeneracy is *at least* $\dim V^{(i)}$. That degeneracy is *enforced* by the symmetry. The level can be more degenerate still, if another representation happens to have the same energy. Lecture 6 makes this precise, including accidental degeneracy and the case of repeated irreducibles.

For example, let an atom sit at the centre of an equilateral triangle (Figure 4.3). Its orbitals $2p_x$ and $2p_y$ transform like the coordinates x and y (Lecture 3), so the rotation r carries $2p_x$ to the superposition

$$\rho(r)\, 2p_x = -\frac{1}{2}\, 2p_x + \frac{\sqrt{3}}{2}\, 2p_y.$$

Let $2p_x$ and $2p_y$ be stationary states with energies E_x and E_y. The Hamiltonian commutes with $\rho(r)$, so $\rho(r)\,2p_x$ is a stationary state with energy E_x. This superposition is a stationary state only if $E_x = E_y$. So the symmetry *forces* the two orbitals to have the same energy.

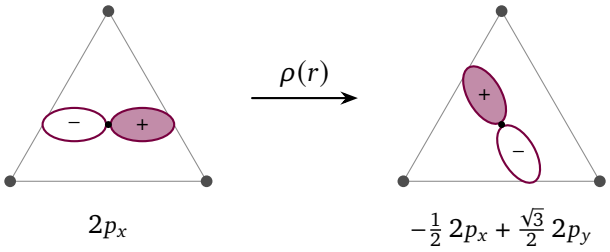


Figure 4.3: Left: the orbital $2p_x$ of an atom at the centre of the triangle. Right: the rotation r turns it through $2\pi/3$, into a superposition of $2p_x$ and $2p_y$.

We can now answer the question this section opened with: how precise is Maschke's decomposition? The subspaces W_j are not unique, but the multiplicities are. Let $V = W_1 \oplus \cdots \oplus W_k$ with every W_j irreducible. The projections onto the summands W_j are intertwiners, because each W_j is invariant. So an intertwiner from $V^{(i)}$ to V splits into one piece for each W_j, and by Corollary 4.14

$$\dim \mathrm{Hom}_G\big(V^{(i)}, V\big) = \sum_{j=1}^{k} \dim \mathrm{Hom}_G\big(V^{(i)}, W_j\big) = m_i.$$

The left-hand side is defined without choosing any decomposition, so every decomposition gives the same multiplicities. Lecture 5 computes them without hunting for invariant subspaces.

Summary.

- *Irreducibility.* An invariant subspace is one preserved by G; a representation is irreducible when it has no proper invariant subspace (Definition 4.1).
- *Averaging.* Every representation of a finite group is unitary for the averaged inner product (Theorem 4.3); the average is invariant because $h \mapsto hg$ only reorders the sum.
- *Maschke.* In a unitary representation the orthogonal complement of an invariant subspace is invariant, so every representation of a finite group is a direct sum of irreducibles, $V \cong \bigoplus_i m_i V^{(i)}$ (Theorem 4.8).
- *Schur's lemma.* An intertwiner of irreducibles is zero or an isomorphism (Theorem 4.12); over $\mathbb{C}$, an operator commuting with an irreducible representation is a scalar (Theorem 4.13). So abelian groups have only one-dimensional irreducibles, and the χ_k are all those of C_n (Corollary 4.15). The multiplicities $m_i = \dim \mathrm{Hom}_G(V^{(i)}, V)$ do not depend on the decomposition.

Tutorial. Exercise Sheet 4.

References

Jones, *Groups, Representations and Physics*, Ch. 3 and §4.1; Tung, *Group Theory in Physics*, §§3.3–3.5; Fulton–Harris, *Representation Theory*, §§1.2, 2.2; Serre, *Linear Representations of Finite Groups*, §§1.3–1.4, 2.2.